

LOVKESH BAROWALIA

Pathankot, Punjab, India

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EDUCATION

Dr. B. R. Ambedkar National Institute of Technology Jalandhar

November 2022 – June 2026

Bachelor of Technology, Information Technology | CGPA: 7.21

Jalandhar, Punjab, India

EXPERIENCE

Bharat Electronics Limited

June 2025 – August 2025

AI/ML Intern | YAMNet, AST, Transfer Learning

Ghaziabad, India

- Engineered an end-to-end acoustic event detection system for defence surveillance, curating a composite dataset of 8,732+ labeled clips across 10 sound classes from UrbanSound8K and AudioSet to classify 3 security-critical event types (vehicles, aircraft, gunfire).
- Implemented an ML pipeline with preprocessing (log-Mel spectrograms) and augmentation (Mixup, SpecAugment), improving model robustness and generalization on noisy real-world audio.
- Applied transfer learning strategies, utilizing YAMNet as a feature extractor and fine-tuning AST, achieving about 93% classification accuracy with high recall for critical events.

PROJECTS

SeatLock | *Node.js, Express.js, PostgreSQL, Docker* | [Link](#) | [GitHub](#)

May 2026 – Present

- Designed a concurrency-safe seat-booking backend using PostgreSQL transactions with row-level locking (SELECT...FOR UPDATE) to eliminate race conditions and guarantee 0 double-bookings.
- Verified architectural correctness under load, ensuring zero double-bookings across **10,000** simultaneous requests with a **~590 bookings/sec** throughput across **1,000** concurrent users.
- Profiled query performance using EXPLAIN ANALYZE, reducing a high-traffic query's execution time by **~95% (9.4ms → 0.1ms)** via targeted secondary indexing.
- Wrote analytics queries using window functions/CTEs and built a Jest/Supertest suite, containerizing the stack with Docker across Render and Neon.

Imagify | *MERN Stack, CLIPDrop API* | [Link](#) | [GitHub](#)

May 2024 – Present

- Architected a production-grade full-stack MERN web application following modular design patterns, successfully deployed and tested by **100+** peer users.
- Engineered secure backend authentication using JWT and bcrypt, implementing custom middleware to protect REST API routes across **2** separate client and server codebases.
- Streamlined core backend image generation and background asset processing to achieve a fast average response time of **5–6 seconds**.
- Automated an application workflow layer featuring credit tracking, tiered subscription plans, and Razorpay integration, hosting live on **2** cloud environments (Vercel and Render).

TECHNICAL SKILLS

Languages: JavaScript, Python, SQL, C++

Backend & Databases: Node.js, Express.js, PostgreSQL, MongoDB (NoSQL), REST APIs, Transactions & Concurrency Control, Docker

Frontend: React.js, HTML, CSS, Tailwind, Three.js

Data and ML: TensorFlow, Scikit-learn, Model Fine-tuning, Data Pipelines

Tools: Linux, Git, GitHub, Postman, Jest, VS Code, Vercel, Render, Neon

Familiar With: CI/CD, AWS, System Design concepts

Practices: Agile Development, Continuous Integration, SDLC

Coursework: Data Structures and Algorithms, Operating Systems, DBMS, Object-Oriented Programming, Machine Learning, Software Project Management, Software Testing, Data Mining and Warehousing.

ACHIEVEMENTS & LEADERSHIP

- Solved **500+** **DSA challenges**, including **400+** **problems** on **LeetCode**, applying core data structures to minimize time and space complexity.
- Served as **Team Captain** of the NITJ Volleyball Club, leading a team of 15+ players to 3 consecutive podiums: Gold at IIT Ropar Aarohan 2026, Bronze at MNIT Jaipur MST 2026 and Aahvaan'26 DTU.
- Organised and led 10+ sporting events for Athletics Fest 2024, coordinating cross-team logistics and driving strong student participation.